Supplementary Material: Majorana zero modes in silicon

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Contents

1 Model, basis, and the physical pairing channel

1.1 Single-channel BdG conventions

The normal-state Hamiltonian on N sites with spacing dx is

$$h = \sum_n c_n^\dagger [(2t - \mu + V_n)\sigma_0 + E_Z \sigma_x] c_n + \sum_n \left[c_{n+1}^\dagger (-t\sigma_0 - i\, a_{so} \sigma_z) c_n + \text{h.c.} \right], \tag{1}$$

with $t = \hbar^2/2m^* dx^2$, $a_{so} = \alpha/2dx$, $E_Z = \frac{1}{2} g \mu_B B$, and singlet pairing $\Delta\, i\sigma_y$. The BdG matrix is assembled as $H = \begin{pmatrix} h & D \\ D^\dagger & -h^* \end{pmatrix}$, $D = \Delta \mathbb{1}_N \otimes i\sigma_y$, which enforces particle–hole symmetry $\mathcal{P} = \tau_x K$ exactly on the lattice. The topological criterion at μ measured from the band bottom is $E_Z^2 > \Delta^2 + \mu^2$. All energies are handled in SI units (J) internally and quoted in μeV.

1.2 Two-valley basis and why pairing is inter-valley

The conduction-band wire has two relevant low-energy valleys at $\pm k_0 \hat{z}$, $k_0 = 0.85\,(2\pi/a_{\text{Si}})$. In the envelope basis (c_+, c_-) (valley Pauli matrices ν_i), time reversal maps $+k_0 \uparrow \to\ -k_0 \downarrow$: *the two valleys are time-reversed partners.* A uniform s-wave parent induces pairing between time-reversed states at zero total momentum; the only zero-momentum singlet channel is therefore *inter*-valley,

$$D \;=\; \Delta\,(\nu_x \otimes i\sigma_y), \qquad D^T = -D. \tag{2}$$

Intra-valley pairing $(c_+ c_+)$ would carry total momentum $2k_0$ (an FFLO-type condensate) and is not induced by a uniform parent; it is kept only in the legacy fig. 4/6 toy model for comparison. Valley splitting enters as valley–orbit (VO) coupling $\lambda(x) = |\lambda(x)| e^{i\phi(x)}$,

$$h_{VO}(x) = |\lambda(x)|\,[\cos\phi(x)\,\nu_x + \sin\phi(x)\,\nu_y] \otimes \sigma_0, \tag{3}$$

with splitting $E_v = 2|\lambda|$; a perfect interface has constant λ, and the phase ϕ is set by the interface position (Sec. 3).

2 The valley equivalence theorem

Theorem 1 (smooth-interface equivalence)**.** *For any* uniform *VO phase ϕ and amplitude $|\lambda|$, the two-valley wire with inter-valley pairing (2), arbitrary valley-scalar terms (kinetic, μ, $V(x)$, Zeeman), and either valley-scalar Rashba SOC ($\alpha k\, \nu_0 \sigma_z$) or phase-locked Dresselhaus SOC ($\alpha k[\cos\phi\,\nu_x + \sin\phi\,\nu_y]\sigma_z$) is unitarily equivalent to two* decoupled *single-band wires with chemical potentials $\mu \mp |\lambda|$, pairing $\pm\Delta$, and the full SOC strength $\pm\alpha$ in each band.*

Proof. Let $U_\phi = e^{-i\phi\nu_z/2} \otimes \sigma_0$ act identically on every site. (i) Valley-scalar terms commute with U_ϕ. (ii) The VO term rotates to $|\lambda|\,\nu_x\sigma_0$ since $U_\phi(\cos\phi\,\nu_x + \sin\phi\,\nu_y)U_\phi^\dagger = \nu_x$. (iii) The pairing block transforms as $D \to U_\phi D\, U_\phi^T$ (not $U^\dagger$ on the right: D couples two creation operators). Because $\nu_z^T = \nu_z$ we have $U_\phi^T = U_\phi$, and using $\nu_x e^{-i\phi\nu_z/2} = e^{+i\phi\nu_z/2}\nu_x$,

$$U_\phi\,(\Delta\nu_x i\sigma_y)\,U_\phi^T = \Delta\, e^{-i\phi\nu_z/2}\nu_x e^{-i\phi\nu_z/2}\, i\sigma_y = \Delta\,\nu_x\, i\sigma_y. \tag{4}$$

The inter-valley pairing is exactly invariant under the valley rotation, for any ϕ. (iv) The phase-locked Dresselhaus SOC rotates, by the same algebra as (ii), to $\alpha k\,\nu_x\sigma_z$. After the rotation everything is diagonal in the ν_x eigenbasis $|\pm\rangle = (|+\rangle_v \pm |-\rangle_v)/\sqrt{2}$: the VO term becomes $\pm|\lambda|$ (a chemical-potential shift $\mu\mp|\lambda|$), the pairing $\Delta\nu_x i\sigma_y \to \pm\Delta\, i\sigma_y$, and the Dresselhaus SOC $\to \pm\alpha k\sigma_z$. The sign of Δ is removed per band by $c \to e^{i\pi/2}c$, and the sign of α by $\sigma_z \to -\sigma_z$ (a spin rotation that leaves $E_Z\sigma_x$ invariant only combined with $\sigma_x \to \sigma_x$; explicitly, the unitary σ_x does this). Hence two decoupled standard wires at $\mu \mp |\lambda|$. □

Numerical verification: `tests.py` (check 3) compares the full $8N$ spectrum against the decoupled-band reference for five phases $\phi \in \{0, 0.7, \pi/2, 2.6, 5.0\}$ and both SOC classes, enforcing agreement to $10^{-6}\,\mu$eV on every eigenvalue (spot checks reach $\sim 3\times 10^{-10}\,\mu$eV, i.e. solver precision). The practical consequence is the wedge picture of Fig. 4 of the main text: at $E_v \gtrsim 100\,\mu$eV one band can be placed in its topological window while the other is trivial, and a smooth interface is benign *regardless of the VO phase value.*

3 Atomic steps

3.1 The step phase $\Delta\varphi = 2k_0(a/4)$

The VO coupling is generated by the sharp interface potential $U_{\rm int}(z)$, which scatters $+k_0 \to -k_0$ with momentum transfer $2k_0$:

$$\lambda \;\propto\; \int dz\, F^2(z)\, U_{\rm int}(z - z_0)\, e^{2ik_0 z} \;\Longrightarrow\; \arg\lambda \;\supset\; 2k_0 z_0, \tag{5}$$

where $F(z)$ is the smooth subband envelope and z_0 the interface height. A single-atomic step on a (001) surface changes z_0 by $a/4$ ($a = 5.431$ Å; the four inequivalent (001) atomic planes per cubic cell are spaced $a/4$). Hence

$$\Delta\varphi = 2k_0\,\frac{a}{4} = 2\cdot 0.85\cdot\frac{2\pi}{a}\cdot\frac{a}{4} = 0.85\,\pi, \tag{6}$$

accidentally close to π because k_0 sits at 85% of the zone boundary in silicon. A double-height (bunched) step gives $2k_0(a/2) = 1.7\pi \equiv -0.3\pi \pmod{2\pi}$, which is *milder* — the basis of the step-bunching design rule in the main text. This is the standard interface-position dependence of valley-orbit coupling [Hosseinkhani–Burkard, PRB 100, 125309 (2019); Losert *et al.*, PRB 108, 125405 (2023)]; what is new here is its consequence under *inter-valley pairing.*

3.2 A step is a Josephson junction: band-projected pairing

Diagonalize the local VO term at fixed x: eigenvectors $|\pm\rangle_\phi = (|+\rangle_v \pm e^{i\phi}|-\rangle_v)/\sqrt{2}$ with energies $\pm|\lambda|$. Projecting the pairing (2) onto the local bands (pairing projects with the *conjugate* right basis, $D_{\rm band} = W^\dagger D W^*$, $W = (|+\rangle_\phi, |-\rangle_\phi)$):

$$\Big(W^\dagger \nu_x W^*\Big)_{\eta\eta'} = \begin{pmatrix} e^{-i\phi} & 0 \\ 0 & -e^{-i\phi}\end{pmatrix}_{\eta\eta'} \;\Longrightarrow\; D_{\rm band} = \pm\,\Delta\, e^{-i\phi(x)}\, i\sigma_y. \tag{7}$$

Each band therefore sees a superconducting order parameter whose phase is minus the local VO phase. A VO phase step $\Delta\varphi$ at x_s is exactly a Josephson junction with phase drop $\Delta\varphi$ in the effective single-band topological wire. The gauge-invariant statement: the relative phase between $\lambda(x)$ and the (uniform) inter-valley pairing cannot be removed — a position-dependent rotation $U_{\phi(x)}$ leaves D invariant [Eq. (4) holds locally] and makes λ uniform, but generates a kinetic gauge field $k \to k - \frac{1}{2}(\partial_x\phi)\nu_z$, i.e. a δ-spike of ν_z-flux at the step; in the band basis ν_z is purely off-diagonal, so the spike both couples the bands and, within each band, reproduces the junction. The two pictures are related by the gauge choice; the spectrum is identical.

At $\Delta\varphi = \pi$ the order parameter changes sign: a topological wire with a π-junction hosts a Kitaev domain wall — an exact zero mode (numerically $E = 0.013\,\mu$eV, Table 1). Near $\Delta\varphi = 0.85\pi$ the bound state is deep. For a transparent short junction the Andreev bound state lies at $E_{ABS} = \Delta_t|\cos(\Delta\varphi/2)|$ with Δ_t the topological gap; Table 1 compares this zero-parameter formula with the lattice numerics.

Table 1: Single-step bound state vs the short-junction Andreev formula ($\Delta_t = 22.22\,\mu$eV, wedge point of Fig. 9; lattice values from `output/data/key_numbers.json`).

$\Delta\varphi/\pi$	0	0.10	0.25	0.40	0.50	0.65	**0.85**	0.95	1.00
lattice E_2 (μeV)	22.22	22.07	19.77	16.08	13.45	9.42	**4.03**	1.34	0.01
$\Delta_t\lvert\cos\frac{\Delta\varphi}{2}\rvert$	22.22	21.95	20.53	17.98	15.71	11.61	**5.19**	1.74	0.00

The formula tracks the numerics to $\lesssim 15\%$ for $\Delta\varphi \lesssim 0.5\pi$ and overestimates by 25–30% (relative to the lattice value) near 0.85π — the expected accuracy for an atomically abrupt junction (zero length but not in the Andreev limit $\Delta \ll$ all normal scales, and the bound state hybridizes with the wire-end MZMs at finite L). We therefore present the junction mechanism as an *analytically derived mapping* [Eq. (7), exact] with a *numerically quantified* bound-state depth, not as a closed-form spectrum.

3.3 Why net winding is irrelevant at realistic step densities

A same-sign staircase accumulates winding $\sum_s \Delta\varphi_s$; a sign-randomized control with identical $|\Delta\varphi|$ accumulates none. Fig. 9(a) of the main text shows the two ensembles are statistically indistinguishable: the damage is per-junction. Winding *does* matter in the smooth limit: a linear ramp $\phi(x) = qx$ is a uniform gauge field, pair-breaking only beyond $q \gtrsim 2|\lambda|/(\hbar v_F)$-type thresholds (in-code generator, `fig9` key numbers: gap reduced by $\lesssim 25\%$ with no subgap state at $q \leq 4\times10^6\,\mathrm{m}^{-1}$, destroyed by 1.6×10^7); at 50 nm step spacing the equivalent $q = 0.85\pi/50\,\mathrm{nm} = 5.3\times10^7$ exceeds this, but the staircase damage at the same mean gradient is dominated by the discrete junction physics, as the control comparison proves.

4 Complete parameter tables

Baseline silicon parameters used throughout unless a figure row overrides them: $m^* = 0.19\,m_e$, $g = 2.0$, $\Delta = 50\,\mu$eV (induced), $dx = 5\,$nm, $\alpha_{\mathrm{demo}} = 0.05\,$eV Å (the "engineered" demonstration value), $\alpha_{\mathrm{Si,opt}} = 10^{-3}\,$eV Å, $\alpha_{\mathrm{InSb}} = 0.5\,$eV Å. Disorder is onsite iid uniform in $[-W, W]$ with correlation length $= dx$ (thresholds are lattice-convention dependent and quoted with dx). Bulk gaps minimize the lower BdG band over $n_k = 6001$ points unless noted. Seeds are `numpy default_rng` with the listed integer families; every random draw in the repository is seeded.

5 Convergence and sensitivity

All entries regenerate from `convergence.py` (`output/data/convergence.json`); Fig. 1 summarizes. The observables are the clean two-valley wedge gap and the single-step (0.85π) bound state at the Fig. 9 baseline ($\mu = 35\,\mu$eV, $B = 1.5\,$T, $E_v = 150\,\mu$eV), plus the hole-platform optimizer outputs.

The one headline number that moves under refinement is the measured-field Si:B center point: $9.9 \to 11.0\,\mu$eV bare ($9.1 \to 9.8$–10.1 renormalized) at finer optimizer grids — the production grid

Table 2: Electron-sector figures. $L = N\,dx$. "IV" = inter-valley pairing model [Eq. (2)]; "toy" = legacy intra-valley pairing.

Fig.	model	N (dx)	μ (μeV)	B (T)	α (eV Å)	valley	disorder	notes
1	1-band	400 (5 nm)	0	0.02–2 (40)	0.05	—	—	validation: fan, MZM, splitting
2	bulk	n_k=6001	−150..150 (201)	0.01–3 (181)	10^{-3} / 0.5	—	—	phase diagrams
3	bulk opt	n_μ=16, n_B=18	0–145	≤ 2	10^{-4}..0.5	—	—	$\Delta(B)$ GL optional
4	toy	200 (5 nm)	−150..150 (31)	1.5	0.05	E_v: 0–300 (31), δ_{iv}=10	—	seed [4-family]
5	1-band	L/dx (case)	0	1.5	case	—	W per case, 16 seeds	cases below
6	toy vs IV	300 (5 nm)	35	1.5	0.05	E_v: 0–300 (13)	—	8 seeds; steps L_s=30,100 nm
7	2D strip	300×5 (5×10 nm)	subband windows	1.5	0.05	—	—	B along wire
9	IV	500 (5 nm)	35	1.5	0.05	$\lvert\lambda\rvert$=75	steps: 6 spacings, 14 seeds	seed family [71,·]
12	RGF	L=2 μm	0	1.5	0.05	—	$W \in \{100, 200, 400, 800\}$	Ando leads, E-scan

Table 3: Fig. 5 disorder cases (per-case dx, L chosen for $k_{so}dx < 0.1$ and $L \gg \xi$ where affordable; 16 realizations each).

case	α (eV Å)	dx, L	W grid (μeV)
intrinsic Si	0.001	5 nm, 20 μm	0, 5, 10, 20, 50, 100, 200, 400
engineered Si	0.05	2.5 nm, 2 μm	0, 20, 50, 100, 200, 400, 800, 1600
strong SOC	0.15	2.5 nm, 6 μm	0, 20, 50, 100, 200, 400, 800, 1600

was ∼10% *conservative*. The Pauli-scenario and Al numbers are grid-exact. Main-text ranges are quoted accordingly.

6 Parent superconductor models and the $g = 2$ optimum

6.1 What Δ is

Δ in the BdG Hamiltonian is a *static induced gap*: the infinite-coupling limit of the standard tunneling self-energy $\Sigma(\omega) = -\gamma\,(\omega + \Delta_p\tau_x)/\sqrt{\Delta_p^2 - \omega^2}$ evaluated at $\omega = 0$, with $\Delta_{ind} = \gamma\Delta_p/(\gamma + \Delta_p)$. In the production figures (i) field suppression enters through the parent, $\Delta_{ind} \leq \Delta_p(B)$, with $\Delta_p(B) = \Delta_0[1 - (B/B_c)^2]$ (GL, orbital-limited parents) or $\Delta_0 - \mu_B B$ (Pauli-limited caricature), and (ii) metallization enters through the quasiparticle weight $Z = 1 - \Delta_{ind}/\Delta_p$, which renormalizes $\alpha \to Z\alpha$ and $g \to Zg + (1 - Z)g_p$ ($g_p = 2$). Both caricatures are stated wherever used. The full dynamical self-energy is solved as a control in Sec. 9.1: it lowers the optimal gaps 24–30% below the static-renormalized values (which therefore act as upper bounds) and agrees with the static model at matched effective coupling to $\lesssim 7\%$ — the caricature's direction and approximate magnitude are confirmed.

6.2 Connection to Cole–Das Sarma–Stanescu

Ref. [Cole *et al.*, PRB 92, 174511 (2015)] showed that strong semiconductor–superconductor coupling buys induced-gap size at the price of semiconductor properties (g, α), so an intermediate coupling optimizes the topological gap. Our $g = 2$ ceiling is the sharp small-g limit of that statement: with $E_Z(2\,\mathrm{T}) = 116\,\mu$eV fixed by $g = 2$, the topological condition $E_Z > \sqrt{\Delta_{ind}^2 + \mu^2}$ cannot be met at all for $\Delta_{ind} \geq 116\,\mu$eV, and maximizing the gap over $(B, \Delta_{ind} \leq \Delta_p(B), \mu)$ under GL suppression yields an interior optimum $\Delta_0 \approx 98\,\mu$eV (main-text Sec. 3.2) — i.e. for silicon electrons the Cole optimum is not merely beneficial but *mandatory*, and it caps the achievable gap at a value disorder then destroys. The hole platform escapes through g_z up to 4.2, which moves the ceiling to $\sim 240\,\mu$eV at 2 T.

Table 4: Hole-sector figures. Parent models in Sec. 6. Optimizer: gap maximized over $B \leq B_{\max}(n_B{=}7)$, $\Delta_{ind} \in [8, \Delta_p(B)]$ ($n_\Delta{=}7$), $\mu \in [0, 160]\,\mu$eV ($n_\mu{=}14$), $n_k{=}2501$ (convergence: Sec. 5).

Fig.	model	parameters
8	bulk opt	(α, g) box: $\alpha \in [0.01, 0.3]$ eV Å (log, 20), $g \in [1.2, 4.0]$ (20); $m^* = 0.25$; center point $(\alpha, g) = (0.06, 2.2)$; parents Si:B-Pauli / Si:B-measured / Al; metallization $Z = 1 - \Delta_{ind}/\Delta_p$, $\alpha \to Z\alpha$, $g \to Zg + (1{-}Z)\, g_{\text{parent}}$, $g_{\text{parent}} = 2$. Finite-wire check at center: $N{=}1200$, $dx{=}2.5$ nm, 10 seeds, $W \leq 800\,\mu$eV.
10	4-band LK	fin $W_y{\times}W_z = 10{\times}12$ nm hard wall, grid $n_y{\times}n_z = 11{\times}13$; $\gamma_1, \gamma_2, \gamma_3 = 4.285, 0.339, 1.446$, $\kappa = -0.42$ (Lawaetz); $E_z \in \{2, 5, 10, 15, 20, 30, 40, 50\}$ MV/m; bare + orbital Zeeman; no strain, no cubic J^3 terms (stated caricature).
11	orientation	$\theta \in [0°, 90°]$ (8), $\phi_B \in [0°, 180°]$ (13); LK tensor at $E_z = 10$ MV/m ($\alpha{=}0.054$, $\lvert g\rvert{=}(0.5\text{–}1.1, g_y, 2.4\text{–}4.2)$, SOC axis $\hat{y}$) vs empirical tensor ($\alpha{=}0.06$, $g{=}(2.1, 2.35, 2.7)$, SOC axis $0.41\hat{y} + 0.91\hat{z}$); parents: thick Si:B (isotropic GL, $B_{c2}{=}0.4$ T) and Al film ($B_{c\parallel}{=}2$ T, $B_{c\perp}{=}0.1$ T); tilted-gap $n_k{=}500$ with the Osca–Ruiz–Serra criterion.
13	convergence	Sec. 5; engine `convergence.py`.

Table 5: Discretization and length convergence (μeV).

	dx (nm), $L{=}2.5\,\mu$m				L (μm), $dx{=}5$ nm		
	10	**5**	2.5		1.25	**2.5**	5.0
clean gap	22.165	22.221	22.234	clean	21.566	22.221	22.239
step 0.85π	4.037	4.034	4.033	step	4.149	4.034	4.034
π-step E_0	0.013	0.013	0.013	ens. median[a]	2.537	0.648	0.083

[a] 50-nm same-sign Poisson staircase, 14 seeds. The L-dependence is physical (Anderson localization; transport section of the main text): ensemble medians are quoted only as L-tagged bounds.

7 Quasiparticle-poisoning estimate

The equilibrium quasiparticle fraction of the parent at temperature T is $x_{qp} \simeq \sqrt{2\pi k_B T/\Delta_p}\, e^{-\Delta_p/k_B T}$. Requiring $x_{qp} \leq 10^{-6}$ with the Si:B parent gap *at the operating field* ($\Delta_p(B^*) = 29$–$33\,\mu$eV for the measured-field design) gives $T_{\text{eff}} \leq 25$–29 mK; the same criterion for the Al reference point of `qp_poisoning.py` ($\Delta_p = 150\,\mu$eV at $B = 1.0$ T, GL) gives 130 mK. Since realistic electron temperatures in superconducting devices rarely reach 25 mK, the measured-field all-silicon stack is classified as a detection platform (ZBP spectroscopy, gap hardness, the miscut test) rather than a parity-coherent qubit, absent parent-gap engineering. All numbers in `qp_poisoning.py`.

8 Luttinger–Kohn extraction

The fin cross-section (y, z) is solved in the 4-band LK Hamiltonian (hard-wall, grid 11×13, $\gamma_1 = 4.285$, $\gamma_2 = 0.339$, $\gamma_3 = 1.446$, $\kappa = -0.42$; parameter provenance: Hensel–Feher cyclotron resonance via Winkler/Venitucci — see `drafts/kp6_benchmarks.md`) with vertical field E_z; the lowest Kramers doublet is followed in k_x to extract m^* (curvature), α (linear-in-k splitting), the SOC axis (spin direction of the split states), and the g-tensor (bare $2\kappa\mu_B \mathbf{B} \cdot \mathbf{J}$ plus orbital terms, evaluated as the field-derivative of the doublet splitting along the three axes). Validation: at $E_z = 0$ the model reproduces the exact bulk [100] heavy/light masses, and the predicted spin–orbit lengths $l_{so} = \hbar^2/(m^*\alpha) = 32$–$101$ nm overlap the measured FinFET window ($\sim$20–60 nm). Limits: no strain, no cubic (J^3) Zeeman, no device electrostatics — the known shortfall is $g_x \in [0.4, 1.1]$ across a seven-geometry bracket (7–16 nm sections, 3–30 MV/m) versus measured $g_{xx} \approx 1.9$–2.3; all platform numbers are therefore carried under both tensors, and the 6-band+Poisson resolution

Table 6: Statistical and parametric sensitivity.

seeds n	7	14	28	56
running median (μeV)	0.724	0.648	0.583	0.638
56-seed bootstrap 95% CI: $[0.516, 0.810]\,\mu$eV				
μ (μeV)	15	25	**35**	45/60
clean / step	9.7 / 1.9	16.7 / 3.0	22.2 / 4.0	24.6–27.6 / 5.0–6.2
E_v (μeV)	100	**150**	225	300
clean / step	11.6 / 2.6	22.2 / 4.0	exits topological window at fixed μ[b]	

[b]At fixed $\mu = 35\,\mu$eV, $B = 1.5\,$T the lower band leaves its topological window for $E_v > 212\,\mu$eV (analytic; grid points $\geq 225\,\mu$eV are outside); single-valley topology at larger E_v requires re-centering μ (main-text Fig. 4 wedge). The quoted values at $E_v \geq 225$ are trivial gaps and are excluded from any protection claim.

Table 7: Hole-platform optimizer convergence and parent-model sensitivity (gap in μeV; (r) = metallization-renormalized).

grid $n_B{\times}n_\Delta{\times}n_\mu$	5×5×10	7×7×14	10×10×20	14×14×28
Si:B measured	8.83	9.92	11.01	11.01
Si:B measured (r)	7.93	9.14	10.13	9.82
Si:B Pauli	30.62	30.62	30.62	30.62
Si:B Pauli (r)	19.68	19.68	19.68	20.21
k-grid n_k: $\leq 0.022\%$ drift from 501 to 12001 at all three operating points.				
Parent model: $\Delta_0 \times \{0.8, 1, 1.2\}$, $B_{c2} \in \{0.3, 0.4\}$ T $\Rightarrow$ measured-field center 8.0–10.6 (7.6–10.0 r).				

is the top open theory item.

9 Parent and disorder realism controls

Four controls (review round 5) test the headline numbers against the caricatures they rest on. All regenerate from `realism.py`, `orbital.py`, `pairing_mix.py`, `morphology.py` (ledgers `realism/orbital/pairing_mi` in `key_numbers.json`).

9.1 Dynamic tunneling self-energy

The static Δ is replaced by the full frequency-dependent BCS self-energy $\Sigma(\omega) = -\Gamma(\omega{+}\Delta_p\tau_x)/\sqrt{\Delta_p^2 - \omega^2}$; quasiparticle energies solve $\det[H(k) + \Sigma(\omega) - \omega] = 0$ (batched eigenvalue crossing scan; α, g, m^* renormalization is automatic through the ω-linear term). Validation: static limit ($\Delta_p \to \infty$, $\Sigma \to -\Gamma\tau_x$) matches `bulk_gap_ueV` to 0.07%; (n_k, n_ω) doubling changes results by 0.001%. The topological criterion is exact at $\omega = 0$: $E_Z^2 > \Gamma^2 + \mu^2$.

Table 8: Dynamic topological gap (μeV) vs tunnel coupling Γ at the three parent operating points ($\mu = 0$, empirical-center hole tensor).

Γ (μeV)	3	6	10	15	20	30	45	70	100	best (Γ^*)	static@$\Delta_{ind}(\Gamma^*)$	static bare
Si:B meas. (0.33 T)	2.7	5.0	**7.6**	3.9	0.6	0	0	0	0	7.6 (10)	7.4	11.0
Si:B Pauli (1 T)	2.7	5.0	7.6	10.4	12.7	**14.9**	7.3	0	0	14.9 (30)	15.2	30.6
Al (1.7 T)	2.7	5.0	7.9	11.1	13.9	18.7	**24.2**	15.3	2.9	24.2 (45)	22.9	50.4

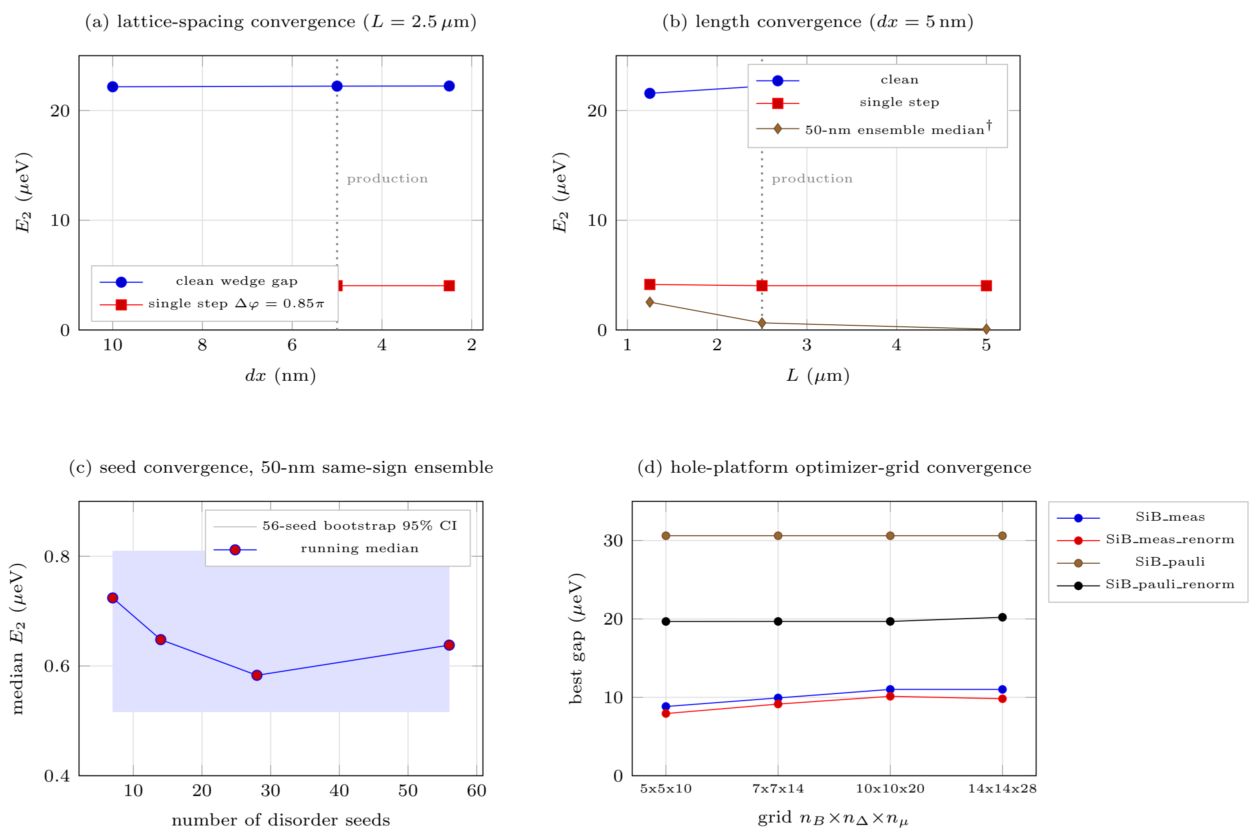

Figure 1: Convergence summary (data: `convergence.json`; rendered by `tools/fig13_pgf.py`, equivalently `convergence.py` with matplotlib). †The L-dependence of the step-ensemble median in panel (b) is physical (Anderson localization), not a discretization artifact; see Table 5, note *a*.

The dynamic optimum is 1.4–2.1× below the static *bare* numbers, sits 24–30% below the static-Z *renormalized* caricature (10.1 / 19.7 / 34.4), and agrees with the static model evaluated at the matched effective induced gap $\Gamma^*\Delta_p/(\Gamma^* + \Delta_p)$ to $\lesssim 7\%$ — the static-Z treatment is directionally validated, and all main-text gap ranges now carry the dynamic correction as the better central estimate. Dynes broadening adds a temperature floor rather than a gap penalty: subgap DOS fraction γ_D/Δ_p dominates the thermal quasiparticle budget below 37–71 mK (Si:B, $\gamma_D/\Delta_p = 10^{-4}$–$10^{-2}$) and 71–136 mK (Al), so 50 mK operation requires $\gamma_D/\Delta_p \lesssim 10^{-3}$ for Si:B parents. Correlated parent-gap inhomogeneity ($\Delta(x)$, 50 nm correlation) is benign at the 10–25% level (median gap 88–97% of clean) and halves the gap at 50%.

9.2 Orbital magnetic coupling

Peierls phases of the perpendicular field component on a 2D strip (validated against `build_wire_2d` to $3 \times 10^{-13}\,\mu$eV at $B_\perp = 0$; exact Hermiticity and PHS). At the preferred tilted Si:B operating point ($B = 0.33$ T, $\theta \approx 39°$, $\Delta_{ind} = 11.3\,\mu$eV) the orbital gap suppression is **0.02–0.04%** for fin widths 7–16 nm and 0.3% at 40 nm (scaling $\propto W^{1.5}B_\perp^2$; the 10% level extrapolates to $W \sim$ several hundred nm). Even a full 1 T out-of-plane field costs only 0.3 / 1.4 / 8.9% at $W = 10/20/40$ nm, with the zero mode and end localization intact. With μ at the lowest subband edge the transverse subband spacing (0.7–6 meV) dwarfs the orbital matrix elements; omitting orbital effects does not alter any orientation-map conclusion at the μeV scale.

9.3 Pairing-channel mixing

$D(\eta) = \Delta[\sqrt{1-\eta^2}\,(\nu_x \otimes i\sigma_y) + \eta\,(\nu_0 \otimes i\sigma_y)]$ interpolates from the physical inter-valley channel to pure intra-valley pairing at preserved total weight ($\eta = 0$ reproduces the production builder to $2 \times 10^{-13}\,\mu$eV). The dilution intuition is *wrong* in the direction favorable to our claims: the valley rotation that removes a VO phase step maps $\nu_0 \otimes i\sigma_y \to e^{-i\phi\nu_z} \otimes i\sigma_y$, so the intra channel inherits the step as a pairing-phase junction — the single-step bound state deepens monotonically ($4.03 \to 3.12 \to 1.68\,\mu$eV at $\eta = 0, 0.1, 0.3$; $2\times$ by $\eta^* \approx 0.24$; near-zero junction modes at $\eta = 1$). Staircase *ensembles* conversely heal with admixture (median E_2: $0.57/1.8/4.5\,\mu$eV at $\eta = 0/0.1/0.3$; the phase-blind intra channel supplies a pairing floor), so the $\eta = 0$ ensemble gaps quoted in the main text are conservative. Physically $\eta \ll \eta^*$: an intra-valley pair carries momentum $2k_0$, which a momentum-conserving uniform parent cannot supply; intra-valley pairing requires interface-roughness Fourier weight at $2k_0$ — the same small parameter that generates the VO coupling itself.

9.4 Step morphology and correlated electrostatic disorder

Morphology (gamma-distributed terraces, shape $k = 4$; miscut θ_m sets the mean terrace $s = (a/4)/\tan\theta_m$; wire at angle χ to step edges; generator validated against the fig. 9 pipeline to $5 \times 10^{-7}\,\mu$eV per seed). Reported as $p_5/p_{50}/p_{95}$ of E_2 (μeV) and the failure fraction $f_{<1} = P(E_2 < 1\,\mu\text{eV})$: miscut scan ($\chi = 90°$): 0.05°: 0.5/2.0/3.1, $f_{<1} = 0.25$; 0.1°: 0.3/1.1/2.8, 0.33; 0.2°: ∼0/0.005/0.02, 1.00; 0.5°: 0/0/0.001, 1.00. Orientation ($\theta_m = 0.1°$): $\chi = 90/45/20/10/5°$ medians 1.5/1.7/2.2/3.2/3.8 with $f_{<1}$ falling $0.33 \to 0$. Finite ramp widths ≤ 10 nm and VO-amplitude dips (to 30% over 3 nm) shift medians by only 10–15% (the latter strictly worsening); double-step bunches ($\Delta\varphi \equiv -0.3\pi$) give 2.7/5.9/10.2 with $f_{<1} = 0$ at the same height budget. Rare-event tails, not medians, are binding: at realistic 0.05–0.1° miscuts 25–33% of realizations fail even where the median looks viable — yield is the constraint. Design rule: wires along step edges ($\chi \lesssim 10°$) on $\leq 0.05°$ substrates, or deliberately step-bunched templates.

Correlated disorder (Gaussian random field $\mu(x)$, correlation length λ_c, vs iid at matched RMS; hole operating point, clean $E_2 = 31.3\,\mu$eV): at $50\,\mu$eV RMS the median collapses to $6.8/4.8/3.5\,\mu$eV for $\lambda_c = 25/50/100$ nm while the iid baseline retains $28.9\,\mu$eV — *the iid model overstates robustness by* 4–8× *in gap at fixed RMS*, making gate-potential smoothness at the few-tens-of-μeV level the binding fabrication requirement. α fluctuations (20%, $\lambda_c = 50$ nm) are negligible (median 29.6); g fluctuations (20%) cost ∼12% (median 27.5). (Caveat: at RMS $= 100\,\mu\text{eV} > E_Z = 64\,\mu$eV the wire fragments and E_2 no longer measures a topological gap; 10 seeds per cell.)

10 Six-band $k \cdot p$ + Poisson: resolving the g_x discrepancy

The decisive theory upgrade demanded in review round 5 (`kp6_holes.py`, `kp6_110.py`, `poisson2d.py`, `kp6_sc.py`; ledger tags `kp6/kp6_110/poisson2d/kp6_sc/ platform110`; literature anchors in `drafts/kp6_benchm`

10.1 Model and validation

Six-band $\Gamma_8 \oplus \Gamma_7$ LKBP Hamiltonian (same γ_i, the standard Chao–Chuang/Winkler coupling blocks, $\Delta_{so} = 44$ meV; Zeeman $H_Z = \mu_B(g_L^h\,\mathbf{B}\cdot\mathbf{L} + g_s^h\,\mathbf{B}\cdot\mathbf{S})$ with $g_s^h = -2$, $g_L^h = 3\kappa + 1$, which reproduces $+2\kappa\mu_B\mathbf{B}\cdot\mathbf{J}$ in Γ_8; identical Peierls orbital treatment as the four-band code). Finite differences do *not* converge for the 6-band problem at affordable grids (the Γ_8–Γ_7 mixing amplifies the $O(d^2)$

error to 45% in α, g_x at 11×13); production uses a hard-wall sine (Galerkin) basis, with the FD mode retained as the conventions bridge. Validation: bulk $k = 0$ exact; [100] HH mass $1/(\gamma_1 - 2\gamma_2)$ to 2×10^{-15}; closed-form bulk branches along [001]/[111] to 3×10^{-15}; $\Delta_{so} \to \infty$ reproduces `lk_holes.extract` to 0.006%; Kramers 2.5×10^{-12}; grid drift $< 3\%$. The [110]-channel rotation (kinetic-tensor rotation, identical pipeline) is validated against the unrotated model at rotated momenta to 10^{-15} and coincides with it in the isotropic $\gamma_2 = \gamma_3$ limit to 10^{-6}.

10.2 Three findings

(i) The split-off band moves g_x the wrong way. At the production fin (10×12 nm, $E_z = 10$ MV/m, [100] channel) the 6-band parameters are $m^* = 0.85$, $\alpha = 0.024$ eV Å, $g = (0.62, 1.03, 4.15)$ vs four-band $(0.44, 0.048, (1.04, 1.99, 3.40))$; the Δ_{so} sweep is monotone (g_x: 1.06 at $\Delta_{so} \to \infty \to$ 0.62 at 44 meV), and the seven-geometry bracket drops to $g_x \in [0.04, 0.93]$. Consistent with Venitucci *et al.*: split-off envelope weight is negligible in the ground doublet; level repulsion makes the channel heavier and transfers g-weight to g_z (which reaches 4.2–4.8, consistent with the published HH-like g_z peak of 4.66–5).

(ii) The discrepancy was predominantly an axes mismatch: all benchmark devices are [110]-channel. Rotating the channel to [110] (the orientation of every measured device — Geyer, Camenzind, Voisin, Crippa) with everything else identical gives a tight, nearly field-independent $g_{x'} = 1.43$–1.66 across the full E_z sweep and all seven geometries, closing $\sim$60% of the gap to the measured along-fin 1.9–2.3; $g_{y'} = 2.0$–2.2 at low field falls inside the measured window, and g_z drops toward the measured 1.5. Favorable side effects: $m^* \approx 0.20$ (vs 0.85), α up to 0.075 eV Å, and the harmful α–g_x covariation of the [100] model largely lifts ($g_{x'}$ varies by only 15% over the whole gate range). The residual 0.3–0.6 shortfall has the size and sign the literature attributes to inhomogeneous electrostatics and 0.1–0.2% process strain.

(iii) Realistic electrostatics is a $\pm 25\%$, geometry-signed correction. The self-consistent tri-gate solution (validated Poisson + SCF, 21 iterations at production parameters; uniform-field control reproduces `extract6` to 0.00%) shifts g_x by -23% (symmetric tri-gate, lateral squeezing) to $+24\%$ (top gate only) at matched density-weighted field; lateral asymmetry ($\Delta V = 0.2$ V between sidewalls) raises g_x by $+31\%$ and α by $+56\%$ together while rotating the SOC axis toward $-\hat{z}$. Electrostatics alone closes at most $\sim$0.1–0.2 of the [100] gap — orientation is the leading correction.

10.3 Platform re-evaluation at [110] parameters

With $(m^*, \alpha, g_{x'}) = (0.19, 0.052, 1.55)$ [$E_z = 10$ MV/m] and $(0.20, 0.073, 1.66)$ [30 MV/m], the *in-plane along-wire* field direction — the only direction film parents tolerate — becomes usable (`platform110`; $10 \times 10 \times 20$ optimizer grids):

Table 9: Hole-platform operating points with six-band [110] parameters. "dyn." = dynamic self-energy optimum (Sec. 9.1 method) at the static operating field; static bare/renormalized as before. In-plane along-wire field except the g_z rows (tilted, thick-Si:B parent only).

scenario	field dir.	static bare	static renorm.	dynamic (Γ^*)
Si:B measured (B_{c2}=0.4 T)	in-plane x'	8.5–8.9	8.1–8.7	3.8 (6)
Si:B measured, tilted	g_z ride	12.5–17.1	10.3–12.6	—
Si:B Pauli film†	in-plane x'	21.0–23.3	16.6–18.7	12.5–13.0 (20)
Al film†	in-plane x'	36.6–40.5	26.0–32.1	16.9–18.4 (30)

Two consequences. First, the film-parent designs no longer require the orientation compromise:

the Al and thin-film Si:B scenarios operate in-plane along the wire at $g_{x'} \approx 1.6$. Second, the honest central estimates (dynamic column) sharpen the tier verdicts: measured-field Si:B falls to $3.8\,\mu$eV — *below* Tier S — so detection-class experiments on the all-silicon stack now require the (hypothetical) Pauli-limited film, while the Al scenario sits at 17–18 μeV dynamic. All numbers remain hypothesis-generating: hard-wall cross-section, no strain, and the [110] tensor still 0.3–0.6 short of measured $g_{x'}$.

11 Reproducibility

Every figure and quoted number regenerates from the committed code: `run_analysis.py` (figures 1–11), `transport.py` / `transport_valley.py` (RGF, invariant, valley transport), `lk_holes.py`, `qp_poisoning.py`, `convergence.py` (Sec. 5), `realism.py`, `orbital.py`, `pairing_mix.py`, `morphology.py` (Sec. 9), `kp6_holes.py`, `kp6_110.py`, `poisson2d.py`, `kp6_sc.py`, `platform110.py` (Sec. 10), `tests.py` (assertion suite covering the equivalence theorem, gauge covariance, analytic dispersions, RGF agreement with closed-chain spectra to 10^{-13}, and regression tests for every bug found in review). The numerical ledger is `output/data/key_numbers.json`; scan checkpoints allow resuming every long run. The repository is public: `https://github.com/wtrevena/feasibility-of-creating-majorana-mode` Archival packaging (commit hash, environment, per-figure commands) is described in `README.md`; randomness is fully seeded (seed families are listed in Tables 2–4).